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A) Current through each resistor
B) Total current in the circuit
C) Total resistance in the circuit

Solution:

A) Since the resistors are connected in parallel, the potential difference across each resistor is same (i.e. V=10V)

Therefore, the current through R_1 is,

$$I_1 = \frac{V}{R_1} = \frac{10}{5} = 2 \text{ A}$$

Current through $R_2 = I_2 = \frac{V}{R_2} = \frac{10}{10} = 1$ A

Current through $R_3 = I_3 = \frac{V}{R_3} = \frac{10}{20} = 0.5$ A

B) Total current in the circuit, $I = I_1 + I_2 + I_3$

$= 2 + 1 + 0.5 = 3.5$ A

C) Total resistance in the circuit $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$= \frac{1}{5} + \frac{1}{10} + \frac{1}{20}$$

$$= \frac{4 + 2 + 1}{20}$$

$$\frac{1}{R_P} = \frac{7}{20}$$

Hence, $R_p = \frac{20}{7} = 2.857\ \Omega$

4. Three resistors of 1 Ω, 2 Ω and 4 Ω are connected in parallel in a circuit. If a 1 Ω resistor draws a current of 1 A, find the current through the other two resistors.

Solution:

$R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$, $R_3 = 4\ \Omega$ Current $I_1 = 1$ A

The potential difference across the 1 Ω resistor $= I_1 R_1 = 1 \times 1 = 1$ V

Since, the resistors are connected in parallel in the circuit, the same potential difference will exist across the other resistors also.

So, the current in the 2 Ω resistor, $\frac{V}{R_2} = \frac{1}{2} = 0.5$ A

Similarly, the current in the 4 Ω resistor,

$$\frac{V}{R_3} = \frac{1}{4} = 0.25 \text{ A}$$

TEXTBOOK EVALUATION

PY4WT

I. Choose the best answer

1. Which of the following is correct?
   a) Rate of change of charge is electrical power.
   b) Rate of change of charge is current.
   c) Rate of change of energy is current.
   d) Rate of change of current is charge.
2. SI unit of resistance is
   a) mho b) joule
   c) ohm d) ohm meter
3. In a simple circuit, why does the bulb glow when you close the switch?
   a) The switch produces electricity.
   b) Closing the switch completes the circuit.
   c) Closing the switch breaks the circuit.
   d) The bulb is getting charged.
4. Kilowatt hour is the unit of
   a) resistivity b) conductivity
   c) electrical energy d) electrical power

10th_Science_Unit-4.indd 55 06-12-2021 18:29:13